Qiaomai Liu

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Summary

I am a research assistant at the School of Chemical Engineering and Technology, Xi'an Jiaotong University, co-advised by Prof. Simin Wang and Prof. Juan Xiao. My recent research work focuses on the experimental and numerical study on the gas combustion of hydrogen blended fuel to mitigate global warming toward world's carbon neutrality by 2050

Education

Xi'an Jiaotong University Xi'an, Shaanxi
M.S. in Power Engineering and Engineering Thermophysics Sep. 2021 – Jun. 2024

- GPA: **3.47/4.00** [87.20/100], Class Rank: 7/21
- Relevant Coursework: Numerical Heat Transfer & Advanced Engineering Thermodynamics

Xinjiang University Urumqi, Xinjiang
B.S. in Process Equipment and Control Engineering Sep. 2017 – Jun. 2021

- GPA: **3.59/5.00** [87.29/100], Class Rank: 4/72
- Relevant Coursework: Process Fluid Machinery & Process Equipment Design

Publications

In Preparation (†: Co-first Author, *: Corresponding Author)

1. **Liu, Q.**, He, S., Xiao, J.*, & Wang, S.* (manuscript drafted). Non-premixed CH_4/H_2-air flame characteristics and pollutant emission with conical bluff body stabilized burner.
2. **Liu, Q.**, He, S., Xiao, J.*, & Wang, S.* (manuscript drafted). Experimental study on combustion characteristics of hydrogen-enriched natural gas in a bluff body-swirl based burner.

Peer-reviewed Publications

3. **Liu, Q.**, He, S., Wang, J., Xiao, J.*, & Wang, S.*. (2023). Effects of Central Split Bluff Body on Combustion Characteristics and NOx Emission with Non-premixed H_2/Air Flames, *Combustion Science and Technology*, 1-27. https://doi.org/10.1080/00102202.2023.2295311.
4. Xiao, J.*†, **Liu, Q.**†, He, S., Wang, S.*, & Zhang Z. (2024). Investigation on the Effects of Hydrogen Addition on Non-premixed Methane/Air Combustion and Emission Characteristics, *International Journal of Hydrogen Energy*, 65, 50-60. https://doi.org/10.1016/j.ijhydene.2024.03.377.
5. Xiao, J., **Liu, Q.**, He, S., Wang, S.*, & Zhang Z. (2024). Prediction and Optimization of Combustion Performance and Emissions for Lean CH_4-H_2 Non-premixed Flame using RSM and PSO, *International Journal of Hydrogen Energy*, 85: 242-251.https://doi.org/10.1016/j.ijhydene.2024.08.234
6. Song, C., **Liu, Q.**, Wang, J.*, Wang, S.*, & Gao, X.* (2024). Estimation of Particle Dispersion Characteristics in a Fluidized Bed with the Binary Mixture of Geldart A and B Types Using the Optical Fiber Probe Method, *Industrial & Engineering Chemistry Research*, 63(17), 7807–7820. https://doi.org/10.1021/acs.iecr.3c04367.
7. He, S., **Liu, Q.**, Wang, S.*, & Xiao, J.* (2023). Two-phase Flow Simulation and Surrogate-Assisted Optimization of Gas Film Drag Reduction in High-concentration Coal-water Slurry Pipelines, *Chinese Journal of Chemical Engineering*, 74(9), 3766-3774. https://doi.org/10.11949/0438-1157.20230674.
8. Duan, X.†, Zhou, A.†, **Liu, Q.**, Xiao, J., & Wen, J.* (2024). Numerical Simulation and Structural Optimization of Electrolyzer Based on the Coupled Model of Electrochemical and Multiphase Flow. *International Journal of Hydrogen Energy*, 49, 604-615. https://doi.org/10.1016/j.ijhydene.2023.08.312.
9. Wang, J.†, Song, C.†, **Liu, Q.**, & Wang, S.* (2024). Pilot-scale Investigation on a Spraying Pretreatment Process Integrated with an Axial Flow Cyclone to Separate Fine Particles with Low Energy Consumption, *Industrial & Engineering Chemistry Research*, 62(39), 16125-16138. https://doi.org/10.1021/acs.iecr.3c02146.
10. Wang, J.†, Deng, Z.†, **Liu, Q.**, & Wang, S.* (2024). Process Characteristics and Energy Efficiency in the Submerged Combustion Treatment of High-salt and High-COD Wastewater, *Case Studies in Thermal Engineering*, 56, 104262. https://doi.org/10.1016/j.csite.2024.104262.
11. Chen, S., Xu, Y., **Liu, Q.**, & Xiao, J.* (accept). Simulation study on the flame morphology and pollution emissions of methane-hydrogen diffusion flames, *Chemical Engineering (China)*.

Patents

12. Zhu, R., Chen, X., Wang, S., Xiao, J., **Liu, Q.**, et al. (review). A Methane-hydrogen Mixing Flow Rate Adaptive Combustion Testing System and Testing Method, Chinese Patent, 2023.

13. Hu, G., Chen, F., Wang, H., Zhang, G., Zhuang, L., Wang, Z., **Liu, Q.**, et al. (review). A High-efficiency and Stable Combustion Testing System and Process for Natural Gas with Hydrogen Addition, Chinese Patent, 2023.

Conference Presentations (underline: presenter)

14. **Liu, Q.**, Xu, S., & Wang, S.* Study on Flame Characteristics and Pollutant Distribution of Methane-Hydrogen Combustion. Chemical Engineering Machinery Conference, Xi'an, China, 24-26 March, 2023. [abstract] [presentation] (in press).
15. **Liu, Q.**, Song, C., Wang, S.*, & Wen, J. Numerical Simulation of Operation and Structural Parameter Optimization for Industrial Riser Reactors. The 11th National Conference of Fluidization and Particle Technology, Hangzhou, China, 17-19 October, 2021 [abstract] [poster].

Grants

Delineated the proposal artworks for the National Natural Science Foundation of China (No. 22308273) [artwork] 2023

Awards & honors

Excellent Young Scholar Paper Presentation Award, Chemical Engineering Machinery Conference	2023
Course Scholarship for Graduate Student, Xi'an Jiaotong University	2021 – 2023
Scholarship for Graduate Students Recommended for Admission, Xi'an Jiaotong University	2021
Direct Admission to Graduate School with Entrance Examination Waived, Minister of Education of China	2020
Scholarship for Motivation, Xinjiang Uygur Autonomous Region People's Government	2018 – 2020
Advanced Individual Award in the Practice of Educating People Thematic Activity, Xinjiang University	2018

Professional experiences

Hydrogen Blended NG Combustion Experiment | *CFD, Python & Experimental devices* Sep. 2022 – Current

- Designed the non-premixed hydrogen blended Natural Gas (NG) combustion system scheme on a laboratory scale; set up and improved experimental platform
- Projected the integration of two novel flame holders (bluff body & swirler) to achieve a highly efficient, safe hydrogen-blended combustion tested with reduced pollutant emissions
- Conducted a numerical investigation on the combustion behavior of hydrogen-blended NG using the EDC model coupled with GRI-Mech 3.0; implemented the physical experiment of the temperature measurement, flame observation, and exhaust gas analysis with Testo 350
- Delineated the poster of "Numerical Simulation and Experimental Research on Combustion" [poster]

Submerged Combustion Experiment | *Inventor, SketchUp & Experimental devices* Aug. 2022 – Jun. 2023

- Drafted the experimental scheme of submerged combustion experiment system by designing the test platform and operation conditions with the full-scale 3D model for visualization and further practical manufacturing
- Improved the submerged combustion system based on the implementation of successive combustion operations and data collection, achieving safety and sustainability
- Identified the experimental temperature measurement problem of multiple thermocouples, influenced by both high-temperature gradients and a harsh two-phase evaporation process under extreme testing conditions

The Cold Dual-Dispersed Fluidization Experiment | *Solidworks & Experimental devices* Aug. 2021 – Jun. 2022

- Prepared the experimental raw materials of Geldart A and B particles; measured the particles' composition and size with the X-ray Diffraction (XRD) and the laser particle size analyzer
- Completed the experimental observation, monitoring, and analytical measurement process of bed pressure drop, particle swelling height, in-bed particle agglomeration concentration with PC6M method
- Solved the obstacle-influenced data measuring process and experimental operation procedure by making the proper geometrical adjustments and physical improvements to the fluidized bed observation window

References

Dr. Zaoxiao Zhang Email: zhangzx@mail.xjtu.edu.cn
Director of the Institute of Chemical Machinery, Professor of Chemical Engineering and Technology School of Xi'an Jiaotong University

Dr. Yunsong Yu Email: cloud.pine02@mail.xjtu.edu.cn
Vice Dean, Professor of Chemical Engineering and Technology School of Xi'an Jiaotong University

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Associate Professor of Chemical Engineering and Technology School of Xi'an Jiaotong University

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